

Assessment - I

V. Bharathsanjay

192525290

MLA0406

- ① A coin is tossed 10 times and the outcomes are: HHTHHHTHHT
Using Maximum Likelihood Estimation, estimate the probability of getting heads.

② Given:

Out come: HHTHHHTHHT

Total toss (n) = 10

No. of Heads (x) = 7

Formula:-

$$\hat{p} = \frac{x}{n}$$

Calculation:

$$\hat{p} = \frac{7}{10} = 0.7$$

$\therefore$ Estimated probability of getting Heads = 0.7 (70%)

- ② out of 50 students, 42 passed an exam

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

③ Given:

Total students (n) = 50

student passed (x) = 42

Formula:- $\hat{p} = \frac{x}{n}$

Calculation:

$$\hat{p} = \frac{42}{50} = 0.84$$

$\therefore$ MLE of passing probability = 0.84 (84%)

- ③ A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

① Given:

Total bulbs (n) = 100

Defective bulbs (x) = 8

Formula:

$$\hat{p} = \frac{x}{n}$$

Calculation:

$$\hat{p} = \frac{8}{100} = 0.08$$

$\therefore$ Probability that a bulb is defective = 0.08 (8%)

- ④ A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling 6 using MLE

① Given,

Total rolls (n) = 60

No. of times 6 appears (x) = 18

Formula:-

$$\hat{p} = \frac{x}{n}$$

Calculation:

$$\hat{p} = \frac{18}{60} = 0.30$$

$\therefore$ The estimated probability of rolling = 0.30 (30%)

5) Given:

$$\text{Input } (x) = 4$$

$$\text{weight } (W) = 2$$

$$\text{Actual output} = 10$$

Procedure:

$$y = W \times x$$

1. Find the predicted output.

2. Calculate the error.

3. If the gradient is 4 and learning rate is 0.01. Find the updated weight.

6) Given,

$$x(\text{Total rolls } n) = 60$$

$$\text{No. of times 6 appears } (x) = 18$$

Formula

$$\hat{p} = \frac{x}{n} = \frac{18}{60} = 0.3 \times$$

Given,

$$\text{input } (x) = 4$$

$$\text{weight } (W) = 2$$

$$\text{Actual output} = 10$$

Predicted formula

$$y = W \times x$$

$$\text{Gradient} = 4$$

$$\text{Learning Rate} = 0.01$$

① Formula

$$y = w \times x$$

$$y = 2 \times 4 = 8$$

$\therefore$ predicted output = 8

② Calculate the error

Formula:-

Error:- actual output - predicted output.